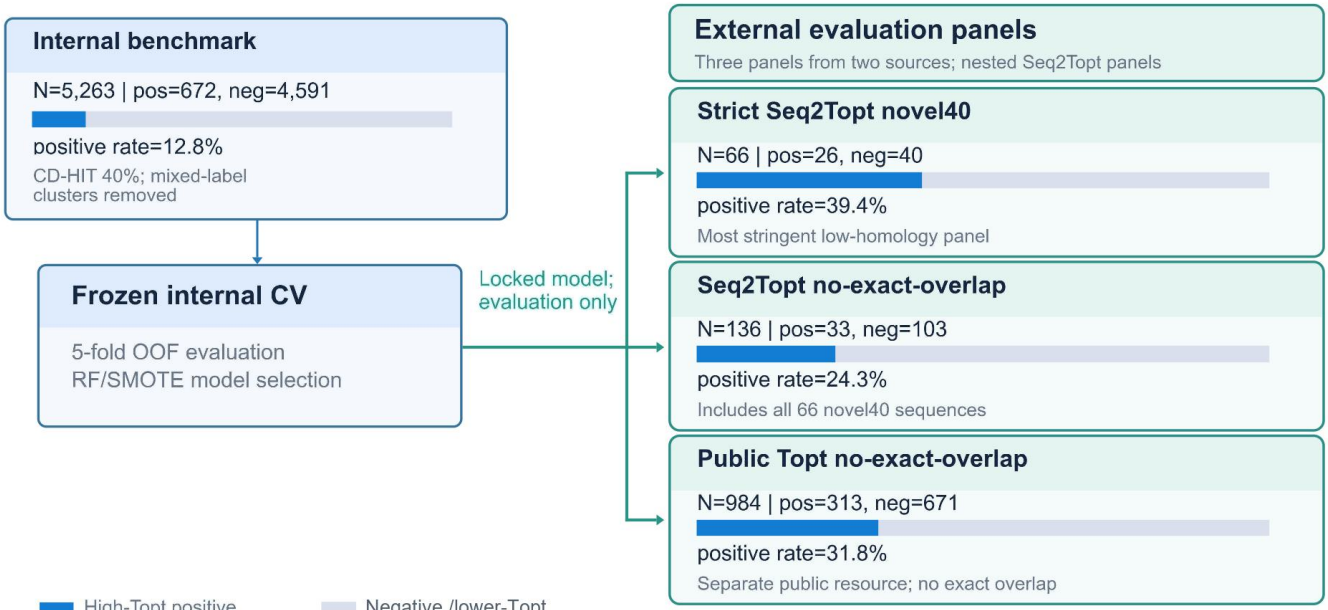

Internal benchmark
N=5,263 | pos=672, neg=4,591
positive rate=12.8%
CD-HIT 40%; mixed-label clusters removed
Frozen internal CV
5-fold OOF evaluation
RF/SMOTE model selection
Locked model; evaluation only
External evaluation panels
Three panels from two sources; nested Seq2Topt panels
Strict Seq2Topt novel40
N=66 | pos=26, neg=40
positive rate=39.4%
Most stringent low-homology panel
Seq2Topt no-exact-overlap
N=136 | pos=33, neg=103
positive rate=24.3%
Includes all 66 novel40 sequences
Public Topt no-exact-overlap
N=984 | pos=313, neg=671
positive rate=31.8%
Separate public resource; no exact overlap
High-Topt positive
Negative /lower-Topt
N=136 and N=984 complement, rather than replace, N=66.
Box area does not encode N; bars show within-panel class balance.